Topological Character Recognition and Semantic Post-Processing for Sinhala Document Accessibility

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Abstract- Sinhala printed document text is often unreadable to blind and visually impaired users, not because recognition fails outright, but because the errors it leaves behind rarely get caught before the text is spoken aloud. This paper tackles both problems on the same Tesseract-based engine. First, focusing on text titles, we test Medial Axis Transform (MAT) skeletonization as a topological preprocessing step against a standard (RAW) pipeline, across a large benchmark, a human perceptual survey, a 135-font stress test, and a geometric complexity analysis. RAW is the stronger generalist (CER 0.0586 vs. 0.0922, $p < 0.001$), but MAT wins on a narrow subset of structurally complex fonts. While MAT incurs a higher preprocessing overhead, utilizing it within a selective hybrid-routing architecture limits the overall system latency increase to under 1%. Second, focusing on body text, a fine-tuned mT5-small corrector cuts character error rate by 36.7% and word error rate by 51.2% on real photographed newsprint (applicable to scanned documents as well), where classical spell-checking baselines fail outright. A 1,329-operation error taxonomy shows dependent vowel signs are the largest and hardest-to-repair category, and that correcting every sentence in full beats gating correction behind an error detector, in every category. Together, the findings point toward a practical hybrid pipeline: a selectively invoked structural preprocessor paired with a full-sequence semantic corrector.

Keywords - Sinhala OCR, Medial Axis Transform (MAT), skeletonization, topological preprocessing, LSTM, post-OCR correction, optical character recognition, mT5, Character Error Rate (CER), assistive technology, accessibility

I. INTRODUCTION

Vision impairment affects a substantial share of the global population, and accessible reading technology remains a persistent gap for many low-resource languages [1], [2] a gap whose reduction is aligned with global commitments to reduce inequality for persons with disabilities [3]. A screen reader assumes the text it is given already exists digitally; a blind reader holding a printed Sinhala newspaper (or any other type of Sinhala article) does not have that prerequisite. The obvious remedy is to photograph the page, recognize the text, and speak it, which depends on two things: recognition must be accurate enough to be intelligible, and residual errors must be caught before reaching a listener who cannot skim back or recover a garbled word visually.

Sinhala presents specific challenges at both stages. It has a large glyph inventory, touching conjunct consonants, and vowel signs (pili) that attach above, below, before, or around a base character [4] and are often only a few pixels tall in photographed newsprint. Sinhala NLP tooling has historically lagged behind higher-resource languages [5], and while recent work applies deep learning to Sinhala OCR [6], benchmarks OCR models for Sinhala and Tamil [7], and builds assistive reading systems on OCR and text-to-speech [8], [9], this work relies on modern feature-learning recognition and, where post-processing exists, rule-based correction rather than a learned language model. Two questions remain open: whether a topological, skeleton-based representation helps a modern OCR engine on a structurally difficult script, and whether a transformer can repair errors left by recognition on real photographed text.

This paper addresses both questions using the same recognition engine. At the recognition stage, we build two parallel Sinhala OCR pipelines identical except for one step: a standard (RAW) pipeline on binarized text-line crops, and a topological (MAT) pipeline adding Medial Axis Transform skeletonization before recognition. We evaluate both across four lines of evidence: a large-scale benchmark, a blind human perceptual survey, an adversarial stress test across 135 difficult fonts, and an objective geometric complexity analysis, alongside each pipeline's computational cost. At the correction stage, we evaluate transformer-based post-OCR correction on a locked, page-disjoint benchmark of 217 sentence pairs from 98 photographed articles, comparing a full-sequence mT5-small corrector, a detector-gated variant, and two classical baselines, with a frequency-weighted error taxonomy and per-category repair analysis.

The contributions are: (i) the first study, to our knowledge, of MAT-based skeletonization as a preprocessing stage for a modern LSTM-based Sinhala OCR engine, validated by quantitative, perceptual, adversarial, and geometric evidence; (ii) the first transformer-based post-OCR correction benchmark for Sinhala on real photographed newsprint, with a frequency-weighted error taxonomy and a quantified negative result for detector-gated correction; and (iii) a discussion situating both interventions within one accessible Sinhala reading pipeline, drawing separate, stage-specific recommendations with a complexity-gated hybrid routing for recognition and a full-sequence over detector-gated

correction, both without assuming the two studies share a common cause.

II. Related Work

A. Topological and Stroke-Based Representations

Thinning and skeletonization are well-established techniques in document image analysis: Lam, Lee, and Suen survey thinning methods for one-pixel-wide skeletal representations of character strokes [10], the topology-preserving approach revisited here. Bag and Harit [11] and Suresh et al. [12] propose related Medial Axis based thinning strategies, but both are algorithmic and script-general (English/Bengali), evaluating neither downstream recognition accuracy on a modern trainable engine nor Sinhala. Rather than proposing a new algorithm, this paper empirically tests whether this established representation helps or hurts a contemporary LSTM-based Sinhala recognizer, validated across four independent methodologies. Cheng et al.'s stroke-based graph-neural sketch representations [13] similarly treat stroke topology as a first-class structural signal, reinforcing that skeleton-level structure can carry information pixel-level features miss.

B. Sinhala OCR and Recognition Engines

Both stages build on the Tesseract OCR engine and its LSTM-based architecture [14], representative of the modern feature-learning era of OCR. Prior Sinhala work includes a survey of NLP tools [5], deep-learning Sinhala OCR [6], an OCR benchmarking study for Sinhala/Tamil [7], a mobile Sinhala book reader for visually impaired students [8], and the SinFUND/SinOCR benchmarks [15], which supply this paper's recognition-stage dataset. Jayatilleke and de Silva report Surya at 2.61% WER against Tesseract 5.5.0 at 14.89% on synthetic Sinhala images [16], noting this understates error on photographed print consistent with the 0.336 raw WER measured here on real photographs. None of this prior work evaluates skeletonization-based preprocessing or transformer-based post-OCR correction, the gap this paper addresses.

C. Post-OCR Correction

Post-OCR correction is established for high-resource languages and studied for several low-resource ones [17], [18]. Published Sinhala work is rule-based: Balasooriya applies Unicode normalization, confusion-pair substitution, and lexicon lookup, raising word accuracy from 53.22% to 86.16% on two single-font pages [19], including the ඒ/ඓ confusion, also the most frequent substitution in this paper's own corpus. Wasala et al.'s n-gram spell checker [20] is implemented here as a classical baseline. Transformer-based Sinhala correction exists but targets a different source: Whisper, SinBERT, and mT5/Mistral correcting dyslexic adults' speech-transcribed writing [21], architecturally similar to the gated variant evaluated here but applied where context usually recovers the intended word. Guan and Greene train ByT5 on synthetic post-OCR data for eight languages [17], and Kashid and Bhattacharyya generate post-OCR data for six Devanagari languages by treating errors as mistranslations [18] , both establishing synthetic noise as a workable substitute for annotated data, the strategy adopted here.

III. Methodology

A. Recognition-Stage Dataset

The primary dataset for the recognition-stage evaluation is SinOCR [15], comprising 100,000 Sinhala text-line images split into a 90,000-image training and 10,000-image test partition. All images were resized to 384px in height, maintaining the original aspect ratio, before processing. Due to few images that failed Tesseract conversion in the RAW pipeline prior to splitting, the resulting length discrepancy (99,798 vs. 99,800 items) desynchronized the deterministic dataset shuffle, shifting 1,108 images across the 90/10 boundary differently between the two datasets. Consequently, all baseline results evaluate solely on the exact intersection of 8,872 test images successfully processed and designated as unseen evaluation data by both pipelines.To evaluate generalization beyond the SinOCR distribution, an independent synthetic stress-test set was constructed.

From an initial pool of roughly 500 internet-sourced Sinhala fonts disjoint from the training data, 135 fonts were manually selected based on severe qualitative divergence from standard typography, such as highly irregular stroke widths and exaggerated lengths. This manual categorization was established independently of the geometric complexity scoring introduced later in Section IV-D. Using this 135-font pool, 10,000 images were generated by rendering random 3–12 word phrases from a Sinhala corpus at a height of 384px.

B. Recognition-Stage Pipeline Architecture

The RAW and MAT pipelines share an identical front end and recognition engine, differing only in a single preprocessing stage, summarized in Table I.

TABLE I. Shared Recognition-Stage Pipeline

Stage	Description
1. Layout detection	PaddleOCR layout-box detection and perspective unwarping
2. Cropping / line detection	CRAFT / PaddleOCR text line and word cropping
3. Binarization	Polarity detection and adaptive/Otsu thresholding
4. Skeletonization (MAT only)	Zhang-Suen thinning via skimage.morphology.skeletonize, then 1 px dilation each side (3 px final stroke width)
5. Recognition	Tesseract LSTM engine (OEM 1); model and PSM differ by stream

In the RAW stream, binarized line crops are passed directly to a Tesseract model (sin_raw) trained exclusively on unmodified binarized Sinhala text images, using page segmentation mode 13 (raw line). In the MAT stream, binarized crops are additionally skeletonized to a one-pixel-wide medial axis using Zhang-Suen thinning as implemented in scikit-image, then dilated by one pixel on each side to yield a standardized three-pixel stroke width. These MAT-processed crops are passed to a separate Tesseract model (sin_Custom), trained exclusively on MAT-skeletonized Sinhala text images, using page segmentation mode 8 (single word). Both models share the same underlying LSTM architecture and differ only in the domain of images they were trained and evaluated on. The two streams are run in parallel on the same input text region, allowing every image to be scored under both conditions.

C. Correction-Stage Corpus and Locked Split

The correction-stage corpus is 1230 close-up photographs of individual articles from three Sinhala daily newspapers (Dinamina, Lankadeepa, Divaina), with the ground truth created by referencing each closeup photograph with the web version E-paper. Each photograph is recognized with Tesseract using the Sinhala trained data through the

pytesseract interface at its default engine and page-segmentation settings. Sentence pairs are formed by aligning recognized text against the transcription at character level; a pair requires at least three alignment anchors and an anchor density of at least 0.10, and each pair is classified by match fraction, length ratio, and Latin-character fraction, giving 1489 correctable pairs (88.4%), 6.5% surplus text, 3.6% truncated, and 1.4% misaligned. Of the 1582, only 1575 fell within a 3–60 word range from the working set, drawn from the transcribed pages. The split is at photograph level, not sentence level, since sentences from one article share font rendering, print quality, lighting, and camera geometry; pages were shuffled with seed 42 and divided 40/15/45 into training, development, and test, with the manifest locked before any modelling. Table II gives the resulting sizes together with raw recognition error per split; the closeness of the three values is evidence that the page-level split is not itself biased.

TABLE II. CORPUS AND LOCKED PAGE-DISJOINT SPLIT

Split	Pages	Sentence pairs	Raw-OCR CER
Train	1094	1187	0.1141
Development	33	178	0.1210
Test (locked)	98	217	0.1197
Total	1225	1582	0.1177

Pages were shuffled with seed 42 and split 40/15/45 at photograph level. The corpus comprises 1230 photographs with manual transcription. The test set contains only real photographed text.

D. Error Taxonomy and Synthetic Noise

The taxonomy used to generate synthetic training noise is measured from the 1365 training and development pairs only. Aligning transcription against recognized text, each non-matching alignment operation is recorded as a substitution (spans of at most three characters), a deletion (at most two), or an insertion (at most two), yielding 192 distinct substitution types, 29 deletion types, and 92 insertion types at an overall character error rate of 0.1092. The three most frequent substitutions are ෂ→ෂ (118), ු→ුෟ (91), and ්‍ර→ර (65); the most frequent deletions are ුෙ (30), ි (25), and ් (18); insertions are dominated by two underscore variants (55 and 53), which are rule-line and margin artifacts rather than character-recognition errors. Training text is 24,212 sentences scraped from three Sinhala news sites, 92.96% of them from one source. After NFC normalization, invisible-character stripping, and deduplication, sentences whose similarity to any held-out transcription reaches 0.70 are removed. Noise is applied per word: one to three operations, of which 40% are drawn from the measured confusion distribution weighted by count and the remainder substitute a vowel sign within its Unicode-defined modifier group, delete a character, or insert a modifier; the fraction of words corrupted is calibrated to the measured real word error rate of 0.273. A further 40% of the corpus is emitted unchanged so the model learns to leave correct text alone, and the 1187 real training pairs are up sampled twentyfold, yielding 36,743 sentence-level pairs and 179,931 span-level examples.

E. Systems Compared

At the recognition stage, RAW and MAT are compared as in Section III-B. At the correction stage, five systems are compared on the locked test set: B1, raw recognized text with no correction; a dictionary baseline that replaces unknown words with the nearest vocabulary entry within edit distance 2, ties broken by frequency; an n-gram baseline, a character trigram model scoring edit-distance-2 candidates by log-probability plus frequency, biased against changing the input; B3, mT5-small fine-tuned end-to-end on (noisy, clean) sentence pairs to generate a corrected sentence from the whole input with no prefix or inline markers; and a proposed detector-gated system, where a SinBERT-large token classifier flags error words and a span-level mT5-small (four words of context each side) replaces each flagged word if its output is at most three tokens, copying unflagged words verbatim. The gated design's rationale is safety: text the detector doesn't flag can't be damaged, which matters when output is spoken to a listener who can't see a plausible-sounding error. Its threshold is chosen by false-positive rate rather than F1 — highest recall at ≤10% correct words flagged (78 dev pairs) — giving 0.30 (recall 0.651, precision 0.946, false-positive rate 0.016)..

F. Evaluation Metrics and Statistical Tests

We assess recognition-stage quality using Character Error Rate (CER), via the Levenshtein edit-distance algorithm. Three statistical tests are used for the recognition-stage results: a paired two-tailed t-test on per-image CER for the baseline evaluation; a two-tailed binomial test against a 50% null for the survey; and a paired two-tailed t-test on per-font average CER for the 135-font stress test.

Correction-stage CER and WER use the same character-level alignment; the cost of a non-matching operation is the larger of the hypothesis and reference symbols it spans, and a sentence's rate is total cost over reference length, averaged over the N = 217 test sentences. Over-correction counts, per sentence, how many already-correct words a corrector destroys. Per-category error analysis assigns every error operation to one of ten categories by Unicode code-point class (Sinhala block U+0D80–U+0DFF, plus punctuation, spacing, Latin/digits, and non-script noise); an error counts as fixed only if every reference position it damaged is correct in the output. Significance is assessed by a paired bootstrap over sentences (10,000 resamples), reporting the 95% CI on the difference in mean CER. Because the gated architecture can only correct what it flags, its fix rate is bounded by detector recall ρ; the reachability ratio $\eta = f^{G}/\rho$, where f^{G} is the fraction of all errors the gated system repairs, decides whether a shortfall lies with the detector or the corrector.

IV. EXPERIMENTS AND RESULTS

A. Recognition-Stage Baseline Evaluation

On the common 8,872-image evaluation subset, the RAW pipeline achieved a mean CER of 0.0586, compared to 0.0922 for the MAT. RAW produced a strictly lower CER on 4,428 images, MAT on 1,584 images, with 2,860 images tied.

TABLE III. BASELINE RECOGNITION EVALUATION, 8,872-IMAGE COMMON SPLIT

Metric	Value
Total images evaluated	8,872
Mean RAW model CER	0.0586
Mean MAT model CER	0.0922
Images where RAW better	4,428 images
Images where MAT better	1,584 images
Images where both Tied	2,860 images
Paired t-test	p = 1.655e-70

B. Human Visual Complexity Survey

To test whether each pipeline's advantage tracks human-perceived difficulty, 404 paired trials (808 images) were sampled, each pairing an image where MAT had the lower CER against one where RAW did; blind evaluators, unaware of which pipeline had won, picked the more complex looking image (ex- Calibri against Time New roman, result in the latter being selected as more complex). They judged the MAT-won image harder in 220 of 404 trials (54.5%), a trend that falls short of significance (two-tailed binomial test, $p = 0.0655$). Table IV's cross-tabulation shows an asymmetry: RAW's CER only doubles (7%→14%) on MAT-favored images, while MAT's CER more than doubles (9%→20%, to a higher ceiling) on RAW-favored images where MAT appears to degrade more severely outside its favorable subset than RAW does outside its own.

TABLE IV. CROSS-TABULATED CER BY PERCEIVED-HARDER IMAGE SET

	RAW-won images judged "harder"	MAT-won images judged "harder"
RAW model CER	7%	14%
MAT model CER	20%	9%

C. Font-Level Stress Test

On the 135-font, 10,000-image synthetic stress test, RAW produced a strictly lower CER on 105 of the 135 fonts, versus 29 fonts for MAT. RAW produced more perfect (zero-error) predictions overall (524 vs. 379) and a wider average win margin (0.1685 vs. 0.0577) on the fonts where each pipeline led. Within the 29 fonts where MAT led, 11% of images were flagged as a catastrophic RAW failure for which the MAT pipeline's prediction remained usable, indicating that MAT's practical value in this subset lies partly in catching specific catastrophic RAW failures rather than in a uniformly lower error rate. A paired, two-tailed t-test on per-font average CER across all 135 fonts yields $p = 0.0214$, confirming the font-level difference is statistically significant.

TABLE V. 135-FONT STRESS TEST SUMMARY

Metric	Value
Total synthetic images	10,000
Fonts tested	135
Fonts where RAW led	105
Fonts where MAT led	29
Perfect predictions, RAW	524
Perfect predictions, MAT	379
Avg. win margin, RAW	0.1685
Avg. win margin, MAT	0.0577
Catastrophic RAW failure flag (within MAT-won fonts)	11%
Paired t-test (per-font CER, all 135 fonts)	$p = 0.0214$

D. Geometric Edge Density Analysis

To test whether the 29 fonts on which MAT leads are objectively more structurally complex, a font-level complexity score was computed for each of the 135 fonts using Otsu binarization followed by Canny edge detection, with complexity defined as edge pixels to total foreground pixels ratio. The 29 MAT-favored fonts had a mean complexity score of 1.59, compared to 1.25 for the remaining 105 RAW-favored fonts, a margin of 21.7%. This is strong, non-circular supporting evidence that MAT's advantage is tied to genuine stroke-geometry complexity rather than an artefact of the evaluation setup, although it measures only whether a font is intrinsically complex and not whether skeletonization reduces that complexity.

TABLE VI. GEOMETRIC COMPLEXITY BY FONT GROUP

Font Group	Mean Complexity Score
ListMAT (29 fonts)	1.59
ListRAW (105 fonts)	1.25
Margin of difference	21.7%

E. Computational Cost of Topological Preprocessing

Average end-to-end processing time was benchmarked to evaluate practical system latency. While MAT's morphological preprocessing (Euclidean distance transform and Zhang-Suen thinning) takes approximately 27 times longer than RAW preprocessing (29.2 ms vs. 1.2 ms), this overhead is negligible compared to the downstream OCR inference bottleneck (~1,168 ms). Total end-to-end latency is 1,169.2 ms for a RAW crop and 1,197.2 ms for a MAT crop, a marginal 2.4% per-crop increase. Under the proposed hybrid architecture, the fast RAW pipeline processes all input by default, selectively invoking the MAT pipeline as a fallback for fonts or predictions flagged as high-complexity or low-confidence. If a routing gate redirects 10% of difficult crops to MAT, the average expected latency increases by merely 2.8 ms per word (a 0.24% system-wide increase). This demonstrates that utilizing MAT as a selective fallback provides robustness on complex layouts with virtually no practical performance penalty.

F. Correction-Stage Accuracy

Table VII reports all five correction-stage systems on the locked test set. Full-sequence correction (B3) reduces character error rate by 36.7% and word error rate by 51.2%. A paired bootstrap gives a 95% confidence interval of [−0.0615, −0.0199] for B3 against raw recognition, [−0.0062, −0.0011] for the gated system against raw recognition, and [+0.0163, +0.0581] for the gated system against B3; the gated system is therefore better than no correction but by an order of magnitude less, and significantly worse than the full-sequence baseline it was designed to improve on. Neither classical baseline solves the problem: dictionary correction leaves CER essentially unchanged (0.1193 vs. 0.1197) while reducing WER by 5.1%, and the trigram checker increases CER by 19.7% and WER by 4.7%. The reason is that recognition errors here are visual rather than phonetic or keyboard confusions, which are the most frequent substitution in this corpus, ච→ච, is optically similar but phonetically and orthographically unrelated, so a candidate generator driven by a lexicon or phonotactics has no reason to prefer it, and both classical baselines also replace correct but low-frequency words with common ones.

TABLE VII. CORRECTION ACCURACY ON THE LOCKED TEST SET (N = 217)

System	Method Type	CER	WER	ΔCER	ΔWER
Raw OCR **(B1)**	none	0.1197	0.3358	-	-
Dictionary + edit dist.	classic	0.1193	0.3187	+0.3%	+5.1%

System	Method Type	CER	WER	ΔCER	ΔWER
N-gram spell checker	classic	0.1432	0.3517	−19.7%	−4.7%
Detector-gated	neural	0.1160	0.2938	+3.1%	+12.5%
Full-sequence mT5 **(B3)**	neural	0.0757	0.1640	+36.7%	+51.2%

Δ is the relative reduction against B1; negative values indicate degradation. Paired bootstrap, 10,000 resamples, 95% CI on ΔCER: B3 vs B1 [−0.0615, −0.0199]; gated vs B1 [−0.0062, −0.0011]; gated vs B3 [+0.0163, +0.0581].

G. Error Repair by Category

Table VIII decomposes the 1,329 correction-stage error operations in the test set. Full-sequence correction repairs 61.8% of them and the gated system repairs 22.5%. Sinhala script errors account for 698 operations (52.5%); punctuation and non-script noise introduced by photographing a printed page, including stray underscores, rule-line fragments, margin marks account for 364 operations (27.4%); multi-character operations spanning both account for the remaining 267 (20.1%). Full-sequence correction repairs 82.4% of the formatting group but only 54.6% of the script group, so a substantial part of the headline 36.7% CER reduction is the removal of recognition artifacts rather than linguistic correction, and 54.6% is the more informative measure of the model's ability on Sinhala text itself. Dependent vowel signs are simultaneously the largest category (299 operations, 22.5% of the total) and the weakest of the major script categories for the full-sequence model (50.8%, against 61.8% for consonants). Increasing the number of photographed pages in the corpus from 38 to 1230 moved CER only from 0.0763 to 0.0757, a difference well inside the bootstrap interval, consistent with the finding below that this category's difficulty is optical rather than statistical.

TABLE VIII. ERROR OPERATIONS BY CATEGORY (N = 1,329)

Category	Errors	B3 fix	Gated fix
Vowel sign	299	50.8%	33.1%
Consonant	275	61.8%	38.2%
Mixed (multi-char.)	267	52.4%	8.6%
Noise artifact	176	92.6%	5.7%
Punctuation	153	81.0%	15.0%
Hal kirima	77	55.8%	41.6%
Independent vowel	41	34.1%	9.8%
Latin / digit	20	35.0%	10.0%
Space	15	40.0%	6.7%
Anusvara	6	33.3%	0.0%
All	1,329	61.8%	22.5%

"Fix" is the proportion of that category's errors fully repaired. The last four categories have fewer than fifty observations and are reported without interpretation.

H. Detector-Gated vs. Full-Sequence Correction

Full-sequence correction destroys 111 of 3,350 already-correct words (3.3%); the gated system destroys 21 (0.6%), so gating is 5.5 times safer on this measure and its design intent is confirmed. It does not follow that the design is justified: full-sequence correction achieves nearly three times the repair rate for roughly three additional destroyed words per hundred, and the alternative delivered to the user is raw recognition at a word error rate of 0.336, so for this application the trade favors the full-sequence model. The obvious explanation for the gated system's shortfall is that its recall of 0.651 is too low; sweeping the detector threshold from 0.1 to 0.9 moves test-set recall from 0.693 to 0.575, but gated CER stays between 0.1158 and 0.1168 across the entire sweep and never approaches 0.0757, showing that detector recall is not the bottleneck. The reachability ratio $\eta = f^{G} / \rho$ (Section III-F) confirms this: even a perfect detector could not close the gap to full-sequence correction, because the gated corrector's four-word context window and per-span decoding are themselves less capable than a corrector that sees the whole sentence.

V. DISCUSSION

Taken At the recognition stage, RAW is the optimal default across standard and adversarial domains, while MAT excels specifically on structurally complex, high-edge-density fonts. Although MAT preprocessing is 27x slower, its end-to-end latency impact is merely ~2.4% per crop. This justifies a confidence- or complexity-gated architecture where MAT serves selectively as a fallback, improving accuracy on difficult layouts with just a 0.24% system-wide latency penalty.

At the correction stage, full-sequence mT5 unambiguously outperforms classical and detector-gated baselines. Crucially, dependent vowel signs constitute the most frequent and difficult errors to fix. When recognition misses a vowel sign, text-only correctors must rely on contextual priors rather than visual evidence, explaining why expanding the training corpus barely improved the Character Error Rate (CER). Therefore, future correction models should utilize byte-level tokenization, canonical decomposition, or image-conditioned correction to recover discarded visual evidence.

While evaluated independently on different datasets (synthetic renders vs. real photographs), these interventions are functionally complementary. Together, they address distinct failure points in complex typography and structural misreads with use of a unified pipeline designed to generate reliable text for visually impaired readers.

A. Limitations

This study has several limitations, at both stages. The 135-font stress test and its geometric analysis are entirely synthetic, so these findings are not yet shown to transfer to noisy, real-world scanned or handwritten documents. The human perceptual survey did not reach conventional significance and should be read as suggestive, not conclusive. Because Tesseract's LSTM operates on full text-line strings rather than isolated characters, the recognition study reports aggregate CER rather than a character-level breakdown of which Sinhala structures drive MAT's advantage. On the correction side, only one recognition engine and one document domain were tested, so generality is unproven; about 93% of the synthetic training text comes from a single news source while test pages come from two others, which makes the task harder rather than easier but still limits in-domain claims; four of the ten error categories have fewer than fifty observations and are reported for completeness rather than interpreted; and results are sensitive to the inference stack with the same checkpoint, same test set, and same generation settings differed by 16% relative CER between two library versions, so future benchmarks should pin and report the inference stack, not just the model. Finally, an earlier version of the pipeline used a similarity gate that silently dropped the hardest 57% of recognition–transcription pairs as "unalignable", inflating the apparent CER improvement fivefold before the bug was found and all affected results discarded, a failure mode that is easy to reproduce and invisible in a published result unless a

benchmark states what fraction of candidate pairs its alignment stage rejects.

VI. Conclusion and Future Work

This paper presented an empirical study of two stages of an accessible Sinhala reading pipeline, evaluated separately. At the recognition stage, a standard pipeline remains the stronger generalist overall, while a topological (MAT) pipeline gives a statistically significant, geometrically-grounded advantage on a narrow, structurally complex subset of fonts at a computational cost that rules out replacing the standard pipeline entirely, favoring a hybrid architecture where MAT is invoked selectively as a fallback. At the correction stage, a fine-tuned mT5-small corrector trained on synthetically noised text reduces CER by 36.7% and WER by 51.2% on real photographs, where two classical baselines give no improvement or degrade results, and a detector-gated architecture, despite being 5.5 times safer against over-correction is outperformed by full-sequence correction in every error category, a gap not attributable to detector recall. These conclusions are reported separately: the recognition study links MAT's advantage to font-level geometric complexity, and the correction study independently identifies dependent vowel signs as its hardest error category to repair, but this paper does not establish a causal link between the two.

Future work follows separately from each half of this study. On the recognition side: a character-level confusion analysis of which Sinhala structures benefit most from skeletonization; a skeleton-derived complexity metric (e.g., branchpoint or junction density) to test directly whether skeletonization reduces geometric complexity rather than only measuring it; validation on real-world scanned or handwritten documents; and implementation of the proposed confidence-gated hybrid routing architecture. On the correction side: byte-level tokenization to target dependent vowel signs directly, and canonical decomposition of the text to convert vowel-sign substitutions into simpler mark deletions. Only once these stage-specific analyses are done will it be possible to test whether the two studies' findings are in fact related, an open question this paper leaves unresolved rather than concluded.

The datasets, source code, and complete training pipelines required to reproduce the custom OCR and NLP models presented in this work are publicly available at the GitHub repository: Janindu-Muthunayaka/R26-DS-002.

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